

Answer Lab: Say It, Support It, Improve It

GROW • Science • 40 minutes

I can give a scientific answer with evidence and reasoning, then improve it after audience response.

Prepare before pupils arrive

Setup: Use the three complete on-screen question/evidence sets or prepare matching printed cards, plus a two-column First answer / Next answer sheet.

Safety: Protect private rehearsal and non-speaking routes. No names, uploads, scores or forced recording.

Equipment: Question cards from forces, space and climate; first/next answer sheet; evidence cards; pencil.

Safety: No forced public speaking or recording. Use an audience of one, pointing, quiet handover or exact adult scribing.

Fallback: Use paper cards, drawing, pointing or spoken response. Preserve the first response beside the next response.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Consolidate forces, space and climate; answer scientific questions.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival • 3 min	Name the three parts of a strong short answer. Explain why the first answer must be kept.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter • 3 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 4 min	claim — my answer reasoning — why the evidence matters revise — make meaning clearer Worked practice answer: 'The rover went farther on the smooth surface.' Keep it visible as the pupil's starting Voice; these practice data are not next week's test. Add worked evidence: smooth 46, 44, 45 cm; fabric 18, 20, 19 cm.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 3 min	Adult Audience says what was heard: 'I heard your claim and the smooth-surface evidence.' The adult asks one check, such as: 'How does friction explain the difference?' The pupil chooses the next move. Keep the first answer, add the pupil's revision beside it, then name exactly what became clearer.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Match each complete question—evidence—reason answer or honest limitation to Forces, Space or Climate. Freeze one answer and its limit, then improve it after an audience check.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 15 min	Choose one embedded question—rover friction, day and night, or extra greenhouse gas and warming—give a first answer, have it received by an adult, then revise or justify keeping it. Record: First answer, adult read-back, pupil-selected next move and improved answer or justified decision to keep it.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Explain how audience response made your Science clearer. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move. The pupil writes what they said and what changed.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: The rover went farther on smooth surface: 44–46 cm versus 18–20 cm, because fabric produced more friction.

Yes. The claim, repeated evidence and friction reasoning all answer the same question.

Misconception repair

Repaired. Scientific quality comes from relevant connected reasoning, not length, volume or visual polish.

Guided checkpoint

Yes. The claim, repeated evidence and friction reasoning all answer the same question. Choose the revision where every part connects to the rover question.

FORCES ANSWER • Why less far on fabric?

FORCES. This is a complete Forces route: the claim answers the rover question, the repeated distances support it and friction explains it. Which topic explains changes in an object's movement?

FORCES LIMIT • one small rover

FORCES. This limitation keeps the Forces conclusion within the model and conditions tested. Which topic used the classroom rover model?

SPACE ANSWER • Why day and night?

SPACE. This is a complete Space route: the model observation supports an explanation using Earth's rotation. Which topic includes Earth, sunlight and rotation?

SPACE LIMIT • simplified lamp and globe

SPACE. This limitation prevents the classroom model being treated as a complete copy or as proof by itself. Which topic used a lamp and globe model?

CLIMATE ANSWER • Why can extra gas warm Earth?

CLIMATE. This is a complete Climate route connecting extra greenhouse gas to the energy mechanism and global-average warming. Which topic includes greenhouse gases, infrared energy and global average temperature?

CLIMATE LIMIT • arrows simplify a complex system

CLIMATE. This limitation keeps the answer accurate without discarding the useful energy mechanism. Which topic used arrows to model infrared energy?

Model limitation

One revised answer shows today's reasoning, not everything the pupil knows. Spoken, pointed, drawn and exactly scribed routes are equivalent.

Independent evidence

First answer, adult read-back, pupil-selected next move and improved answer or justified decision to keep it.

Supported: Choose one embedded topic answer, identify its claim, evidence and reason, then communicate it in a preferred mode. Adult reads back exact choices; pupil selects Add evidence, Clarify reason or Keep and explain.

Standard: Answer one of the three embedded questions independently, receive one audience check and revise one part. Use Claim, Evidence and Reason headings only if needed; fade them after the first success.

Stretch: Answer a second embedded topic question, compare evidence strength and justify the revision or no-change decision. Add one limitation without weakening a secure conclusion.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_G_W12B_Science_Answer_Lab_Do.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.